ARRAY PROBLEMS

1.CALCULATE THE SUM OF ALL ELEMENTS IN AN ARRAY.

```
import java.util.Scanner;
public class sum{
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);
      int[] arr = new int[5];
      int sum = 0;
      for (int i = 0; i < 5; i++) {
         arr[i] = sc.nextInt();
      }
      for (int i = 0; i < 5; i++) {
         sum += arr[i];
      }
      System.out.println("Sum = " + sum);
      sc.close();
   }
}
```

2.CALCULATE THE AVERAGE OF ALL ELEMENTS IN ARRAY.

```
import java.util.Scanner;
public class average{
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);
      int[] arr = new int[5];
      int sum = 0;
      for (int i = 0; i < 5; i++) {
         arr[i] = sc.nextInt();
         sum += arr[i];
      }
      double average = (double) sum / 5;
      System.out.println("Average = " + average);
      sc.close();
   }
}
```

3.FIND THE LARGEST ELEMENT IN AN ARRAY

```
import java.util.Scanner;
public class largest {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);
      int[] arr = new int[5];
      for (int i = 0; i < 5; i++) {
         arr[i] = sc.nextInt();
      }
```

```
    int largest = arr[0];
    for (int i = 1; i < 5; i++) {
      if (arr[i] > largest) {
        largest = arr[i];
      }
    }
    System.out.println("Largest = " + largest);
    sc.close();
  }
}
```

4.FIND THE SMALLEST ELEMENT IN AN ARRAY.

```
import java.util.Scanner;
public class smallest {
  public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);
    int[] arr = new int[5];
    for (int i = 0; i < 5; i++) {
      arr[i] = sc.nextInt();
    }
    int smallest = arr[0];
    for (int i = 1; i < 5; i++) {
      if (arr[i] < smallest) {
        smallest = arr[i];
      }
    }
    System.out.println("Smallest = " + smallest);
    sc.close();
  }
}
```

5.COUNT HOW MANY ODD AND EVEN NUMBERS ARE PRESENT IN AN ARRAY

```
import java.util.Scanner;
public class oddeven{
  public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);
    int[] arr = new int[6];
    int evenCount = 0;
    int oddCount = 0;
    for (int i = 0; i < 6; i++) {
      arr[i] = sc.nextInt();
      if (arr[i] % 2 == 0) {
        evenCount++;
      } else {
        oddCount++;
      }
    }
```

```
    System.out.println("Even count = " + evenCount);
    System.out.println("Odd count = " + oddCount);
    sc.close();
  }
}
```

6.SEARCH FOR A GIVEN ELEMENT IN AN ARRAY USING LINEAR SEARCH

```
import java.util.Scanner;
public class search {
  public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);
    int[] arr = new int[5];
    for (int i = 0; i < 5; i++) {
      arr[i] = sc.nextInt();
    }
    int search = sc.nextInt();
    boolean found = false;
    for (int i = 0; i < 5; i++) {
      if (arr[i] == search) {
        System.out.println("Element found at index " + i);
        found = true;
        break;
      }
    }
    if (!found) {
      System.out.println("Element not found");
    }
    sc.close();
  }
}
```

```
import java.util.Scanner;

public class count{
  public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);

    int[] arr = new int[7];
    int positive = 0;
    int negative = 0;
    int zeros = 0;
    for (int i = 0; i < 7; i++) {
      arr[i] = sc.nextInt();
```

```
            if (arr[i] > 0) {
                positive++;
            } else if (arr[i] < 0) {
                negative++;
            } else {
                zeros++;
            }
        }

        System.out.println("Positive = " + positive);
        System.out.println("Negative = " + negative);
        System.out.println("Zeros = " + zeros);

        sc.close();
    }
}
```